

HRISHIRAJ MANDAL

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EDUCATION

National University of Singapore (NUS)

August 2023 - May 2027

Bachelor's, Computer Science

- **Awards:** Winner Ripple X NUS Fintech Summit Hackathon, Nels Radley Freits Smartkarma Fellowship
- Languages: Python, Java, JavaScript, Typescript, Solidity, C, C++, Scheme, OCaml
- Core Modules: Software Engineering, Algorithms and Data Structures, Information Security, Computer Networking, Database Systems, Programming Language Development, AI/ML, Operating Systems, Formal Verification
- Semester projects: Microcontroller development using C, Arduino and Fusion360

PROFESSIONAL EXPERIENCE

University of Birmingham

June 2026 - Present

Formal Verification Research Intern

Birmingham, England, United Kingdom

- Awarded AI Safety Scholarship for outstanding students
- Architected reusable Rust abstraction for CubeSat Space Protocol (CSP) Layer 2 components by applying the adapter pattern to separate shared control flow from protocol-specific implementations to improving memory and temporal safety over the existing C codebase
- Guaranteed correctness and liveness properties of CSP through formal verification workflow using Aeneas and Lean, translating Rust implementations into functional models and representing control flow as Büchi automata

National University of Singapore (NUS)

May 2025 - Present

Teaching Assistant

Singapore

- Research Assistant, ACSL-SL conversion for Frama-C (Dec – Jan 2026): Designed and implemented a specification compiler in OCaml translating Separation Logic to ACSL for Frama-C, enabling automated formal verification of low-level C programs
- Teaching Assistant, Introduction to Information Security (AY25/26): Conducted weekly tutorials for 40 students, explaining cryptography, access control, and vulnerability analysis; supported student consultations
- Teaching Assistant, Independent Software Development Project (AY24/25, 25/26): Mentored 13 student teams through end-to-end software projects (requirements, design, implementation, evaluation), leading 1 team to top 5% of the cohort

The Agency for Science, Technology and Research

January 2025 - July 2025

AI Research Intern

Singapore

- ARIA awardee - Contributed research on 'Demand Forecasting Frameworks for Fast Moving Consumer Goods (FMCG)'
- Designed and evaluated forecasting pipelines using machine-learning time-series models within the Nixtla ecosystem
- Lead development of internal Explainable AI (XAI) package for demand-forecasting models, enabling feature-level and temporal attribution through collaborated in an agile, Git-based research workflow and presentation to cross-functional stakeholders
- Link to testimonial: https://www.hrishiraj.site/assets/astar_testimonial%20and%20cert-BZDhLwBy.pdf

Smartkarma

February 2023 - July 2023

Data Analyst Intern

Singapore

- Streamlined financial data collection pipelines using the Bloomberg API and optimizing data-driven reporting using PostgreSQL
- Testimonial of positive performance evaluation: https://www.hrishiraj.site/assets/SK-general-2xJmd_RW.pdf

PROJECTS & OUTSIDE EXPERIENCE

Oma

May 2024 - August 2024

Backend, Quality Assurance, DevOps lead

Singapore

- Native recommendation app that suggests dining places using AI to learn your preference
- Recommendation algorithm built using PyTorch, Backend using NEXT.JS and TRPC on Turborepo, Quality Assurance using ViTest testing framework and hosted on Microsoft Azure
- [Link to Project](#)

BARE

November 2023 - December 2023

Team Lead

Singapore

- Winning Submission For Ripple, NUS Fintech Summit 2024 Hackathon - rental dApp through Ethereum
- Usage of Solidity to prepare contracts and React.js to craft user interface
- [Link to Project](#)

SKILLS

Skills: Python, Java, JavaScript, TypeScript, Solidity, C#, Pytorch, Next.js, Prisma, MongoDB, Postgres, Excel/Numbers/Sheets, Zapier, SQL, Playwright, Microsoft Azure, Blockchain, React.js, Data Structures & Algorithms, Computer Networking, Data Analysis, Machine Learning, Quality Assurance (QA), DevOps, Market Research, Lean, Rust, Formal Verification

Languages: English (Native), Bengali (Professional), Chinese (Elementary)